Notes: Gul, Kim & Qiu (JFE, 2010)

Ownership Concentration, Foreign Shareholding, Audit Quality, and Stock Price Synchronicity: Evidence from China

Contents

1 Big picture

- **Question.** How do firm-level governance mechanisms affect the amount of firm-specific information incorporated into stock prices in China's emerging equity market?
- **Main outcome.** The paper uses stock price synchronicity, SYNCH, as an inverse measure of firm-specific information capitalization. High synchronicity means returns move mainly with common market or industry factors; low synchronicity means more firm-specific information is reflected in prices.
- **Institutional setting.** China is a useful setting because listed firms have concentrated ownership, many largest shareholders are government-related, some firms issue foreign shares, and Big 4 audits are uncommon.
- **Main ownership result.** Synchronicity is a concave function of largest-shareholder ownership. It rises as ownership concentration rises initially, peaks around a 50% ownership level, and then falls at higher ownership levels.
- **Interpretation of concavity.** At lower ownership levels, entrenchment dominates: large shareholders may hide firm-specific information to extract private benefits. At higher ownership levels, incentive alignment and reputation dominate: large shareholders have stronger incentives to reveal information and protect value.
- **State ownership result.** Stock prices are more synchronous when the largest shareholder is government-related, consistent with weaker investor protection and less firm-specific information disclosure.
- **Foreign ownership result.** Firms with foreign shares have lower synchronicity. The effect is stronger for H-share firms traded in Hong Kong than for B-share firms traded in mainland China, consistent with stronger institutional protections and disclosure in Hong Kong.
- **Audit quality result.** Big 4 auditors are associated with lower synchronicity, suggesting that higher-quality audits improve credibility and facilitate incorporation of firm-specific information into prices.
- **Validation result.** Earnings information is less strongly reflected in returns for high-synchronicity firms, supporting the interpretation of synchronicity as a measure of weak firm-specific information capitalization.
- **Policy takeaway.** In transitional markets, improving firm-level governance, reducing state ownership influence, encouraging credible foreign participation, and strengthening audit quality can make stock prices more informative.

2 Incremental contribution of the paper

2.1 Contribution relative to country-level synchronicity research

- Earlier work documents that emerging markets have high stock price synchronicity and links this to weak investor protection and poor transparency at the country level.
- This paper moves the analysis from the country level to the **firm level** inside one emerging market.
- By holding country-level legal and institutional environment constant, the paper isolates how within-country variation in ownership, foreign participation, and audit quality affects informativeness of stock prices.

Interpretation: The incremental step is important because China is uniformly classified as an emerging market, yet firms within China differ sharply in governance. The paper shows that firm-level governance still matters even in a weak country-level investor-protection environment.

2.2 Contribution relative to ownership concentration research

- Prior governance literature debates whether concentrated ownership helps or hurts outside investors.

- This paper shows that the effect is nonmonotonic: concentration initially worsens the information environment but improves it after a threshold.
- The empirical concavity maps naturally to the tension between managerial entrenchment and incentive alignment.

Interpretation: The paper's incremental contribution is not simply saying that large shareholders matter. It shows that their effect depends on the level of ownership concentration and on whether the largest shareholder is government-related.

2.3 Contribution relative to foreign ownership literature

- Foreign investors can improve information environments through monitoring, information processing, and demand for disclosure.
- The paper exploits China's split-share institutions: A-share firms, A+B-share firms, and A+H-share firms.
- This lets the authors compare foreign ownership in a mainland Chinese trading environment with foreign ownership linked to Hong Kong's more developed market institutions.

Interpretation: The incremental contribution is to show that foreign ownership is not homogeneous. Foreign participation appears most valuable when embedded in stronger market institutions.

2.4 Contribution relative to audit-quality literature

- Audit-quality studies usually focus on earnings management, financial reporting quality, or cost of capital.
- This paper links auditor quality directly to stock price synchronicity, a market-based measure of firm-specific information incorporation.
- The Chinese setting is useful because Big 4 audits are relatively rare, making the contrast between Big 4 and non-Big 4 audits economically meaningful.

Interpretation: The paper adds auditors to the list of governance mechanisms that shape price informativeness. It treats audit quality as part of the information-production system of capital markets.

3 Institutional background and hypotheses

3.1 China's ownership structure

- Chinese listed firms often originate from partial privatizations of state-owned enterprises.
- The largest shareholder typically owns a large block and often has government-related identity.
- Ownership concentration can protect minority shareholders if it aligns incentives, but it can also entrench controlling shareholders and encourage tunneling or selective disclosure.

Interpretation: The paper's first hypothesis is inherently two-sided: concentrated ownership can either suppress or reveal firm-specific information depending on which force dominates.

3.2 Hypothesis H1a: concavity in ownership concentration

The paper tests whether synchronicity is concave in largest-shareholder ownership:

$$\mathrm{SYNCH}_{it} = \gamma_0 + \gamma_1 \mathrm{TOPHOLD}_{it}^2 + \gamma_2 \mathrm{TOPHOLD}_{it} + \gamma_3 \mathrm{TOPGOV}_{it} + Controls + FE + \varepsilon_{it}. \tag{1}$$

The concavity prediction is:

$$\gamma_1 < 0, \qquad \gamma_2 > 0. \tag{2}$$

Interpretation: At low ownership levels, increasing the largest shareholder's stake strengthens entrenchment and reduces information flow. At high ownership levels, incentive alignment and reputation concerns dominate.

3.3 Hypothesis H1b: government-related largest shareholder

- TOPGOV equals one when the largest shareholder is government-related.
- Government-related owners may pursue political or social objectives, face weaker market discipline, and disclose less firm-specific information.
- The hypothesis is $\gamma_3 > 0$: government-related largest shareholders increase synchronicity.

Interpretation: Higher synchronicity under government ownership means stock prices reveal less firm-specific information.

3.4 Hypotheses H2a and H2b: foreign ownership

The foreign-ownership specification is:

$$\text{SYNCH}_{it} = \phi_0 + \phi_1\text{HSHARE}_{it} + \phi_2\text{BSHARE}_{it} + Controls + FE + \varepsilon_{it}. \tag{3}$$

- H2a predicts $\phi_1 < 0$ and $\phi_2 < 0$: foreign shareholding reduces synchronicity.
- H2b predicts the H-share effect is stronger than the B-share effect because H-shares trade in Hong Kong, where market institutions and disclosure rules are stronger.

Interpretation: Foreign investors facilitate capitalization of firm-specific information, but the effect depends on the quality of the surrounding institutional infrastructure.

3.5 Hypothesis H3: audit quality

The audit-quality specification is:

$$\text{SYNCH}_{it} = \lambda_0 + \lambda_1\text{BIG4}_{it} + \lambda_2 LOCAL_{it} + Controls + FE + \varepsilon_{it}. \tag{4}$$

- BIG4 equals one if the firm uses an international Big 4 auditor.
- $LOCAL$ equals one if the auditor is local to the firm's administrative region.
- H3 predicts $\lambda_1 < 0$: Big 4 audits reduce synchronicity.

Interpretation: Big 4 auditors improve credibility of financial reporting and therefore help investors impound firm-specific information into prices.

4 Data and measurement

4.1 Sample

- The sample covers nonfinancial Chinese listed firms from 1996 to 2003.
- Stock return and accounting data come from CSMAR.
- Ownership data are hand-collected from annual reports and other sources.
- Auditor data come from CSRC's *Who Audits China* and supplemental manual collection.
- The final sample has 6,120 firm-year observations for 1,142 firms.

Interpretation: The hand-collected ownership and auditor data allow firm-level governance measurement that standard stock-return datasets do not contain.

4.2 Stock price synchronicity

For all firms, the authors estimate a daily-return market model:

$$RET_{it} = a + b_1 MKTRET_t + b_2 MKTRET_{t-1} + b_3 INDRET_t + b_4 INDRET_{t-1} + e_{it}. \tag{1}$$

For A-share only firms, A+B-share firms, and A+H-share firms, the authors also estimate models including world, B-share, or Hong Kong market returns:

$$RET_{it} = a + b_1 MKTRET_t + b_2 WRDRET_t + e_{it}, \tag{1a}$$
$$RET_{it} = a + b_1 MKTRET_t + b_2 MKTRET_t^B + b_3 WRDRET_t + e_{it}, \tag{1b}$$
$$RET_{it} = a + b_1 MKTRET_t + b_2 MKTRET_t^H + b_3 WRDRET_t + e_{it}. \tag{1c}$$

Synchronicity is the logistic transformation of the market-model R^2:

$$\text{SYNCH}_i = \log\left(\frac{R_i^2}{1 - R_i^2}\right). \tag{5}$$

Interpretation: A higher R^2 means common factors explain more of the return variation. After log transformation, higher SYNCH means less firm-specific information in prices.

4.3 Main variables

- TOPHOLD: percentage of shares held by the largest shareholder.
- TOPGOV: one if the largest shareholder is government-related.
- HSHARE: one if the firm issues H-shares.
- BSHARE: one if the firm issues B-shares.
- BIG4: one if the firm uses an international Big 4 auditor.
- $LOCAL$: one if the firm's auditor is located in the same administrative region.
- Controls include trading volume, firm size, leverage, earnings volatility, market-to-book, number of firms in the industry, and industry size.

Interpretation: The controls absorb liquidity, size, risk, growth opportunities, and industry structure that can independently affect return comovement.

5 Identification and empirical design

5.1 Identification logic

- The paper does not use a randomized or quasi-random shock.
- Identification comes from cross-sectional and time-series variation in firm-level governance characteristics within one country.
- Year and industry fixed effects absorb common macro and sectoral determinants of synchronicity.
- Robust standard errors are adjusted for firm clustering.

Interpretation: The empirical strategy is best interpreted as a carefully controlled association study with strong institutional motivation rather than a causal natural experiment.

5.2 Endogeneity of auditor choice

Auditor choice may be endogenous. The paper estimates a first-stage Big 4 auditor choice model:

$$Pr(\mathrm{BIG4}_{it}) = \delta_0 + \delta_1 LEV_{it} + \delta_2 M/B_{it} + \delta_3 SALES_{it} + \delta_4 SHRINC_{it} + \delta_5 LOSS_{it} + FE + \varepsilon_{it}. \tag{5}$$

It then uses fitted Big 4 probability, $PredBIG4$, and inverse Mills ratio corrections in second-stage synchronicity regressions.

Interpretation: These checks address the concern that firms choosing Big 4 auditors may already have lower synchronicity for other reasons.

5.3 Additional validity test: earnings information and returns

To validate the interpretation of synchronicity, the paper estimates earnings-response regressions:

$$MAR_{it} = \alpha_0 + \alpha_1 NI_{it} + \alpha_2 NI_{it} \times DR\_SYNCH_{it} + Controls + \varepsilon_{it}. \tag{6}$$

- MAR is market-adjusted return.
- NI is earnings scaled by market value.
- DR_SYNCH is a decile-rank measure of synchronicity.

Interpretation: If high synchronicity means less firm-specific information in prices, earnings should be less strongly capitalized into returns when SYNCH is high. Table 8 confirms this.

6 Tables and figures

6.1 Tables 1–2: sample and descriptive statistics

Read:

- Table 1 reports that the sample has 6,120 firm-year observations from 1,142 firms.
- Manufacturing accounts for 58.15% of the sample.
- The number of sample observations rises over time as China's stock market expands.
- Table 2 shows mean $R^2(1) = 0.454$ and mean SYNCH(1) $= -0.232$.
- The average largest shareholder owns 42.8% of shares, and 66.5% of largest shareholders are government-related.
- Foreign ownership is uncommon: H-share firms are 4.0% of observations and B-share firms are 12.3%.
- Big 4 audits occur in only 7.3% of firm-years.

Economic message: China's market has high synchronicity, highly concentrated ownership, extensive state involvement, limited foreign participation, and rare Big 4 audits.

Table 1
Sample distribution.
Panel A shows the distribution of sample firms across industries based on the “guidance on the industry category of listed companies” issued by the China Securities Regulatory Commission (CSRC), where A=Agriculture, B=Mining, C=Manufacturing, D=Electricity, gas, and water, E=Building and construction, F=Transportation and logistics, G=Information technology, H=Commerce, I=Real estate, J=Service, K=Culture and media, L=Conglomerate. Panel B shows the distribution by year.

Panel A: Industry distribution	A	B	C	D	E	F	G	H	I	J	K	L	Total
#	30	15	664	43	18	47	57	99	34	39	10	86	1142
%	2.63	1.31	58.15	3.77	1.58	4.12	4.99	8.67	2.98	3.42	0.88	7.53	100

Panel B: Yearly distribution	1996	1997	1998	1999	2000	2001	2002	2003	Total
#	298	341	538	747	867	972	1140	1217	6120
%	4.87	5.57	8.79	12.21	14.17	15.88	18.63	19.89	100

Table 1: sample distribution

$R^2(1)$ and *SYNCH*(1) refer to the R^2 statistic and the stock price synchronicity measures, respectively, that are estimated using Eq. (1), while $R^2(1a, b, c)$ and *SYNCH*(1*a*, *b*, *c*) refer to the same measures that are estimated using Eqs. (1a), (1b), and (1c) for firms with A-shares only, firms with A- and B-shares, and firms with A- and H-shares, respectively. All other variables are as defined in the Appendix A.

Variables	Mean	Std. dev.	5th Pctl.	25th Pctl.	Median	75th Pctl.	95th Pctl.
$R^2(1)$	0.454	0.180	0.145	0.327	0.462	0.588	0.741
$R^2(1a, b, c)$	0.437	0.179	0.134	0.307	0.439	0.570	0.734
SYNCH(1)	−0.232	0.841	−1.776	−0.724	−0.151	0.354	1.053
SYNCH(1*a*, *b*, *c*)	−0.316	0.882	−1.862	−0.815	−0.247	0.281	1.016
TOPHOLD	0.428	0.176	0.162	0.289	0.417	0.571	0.718
STATE	0.316	0.261	0.000	0.000	0.332	0.542	0.711
TOPGOV	0.665	0.472	0.000	0.000	1.000	1.000	1.000
FOREIGN	0.038	0.104	0.000	0.000	0.000	0.000	0.318
HSHARE	0.040	0.195	0.000	0.000	0.000	0.000	0.000
BSHARE	0.123	0.328	0.000	0.000	0.000	0.000	1.000
BIG4	0.073	0.260	0.000	0.000	0.000	0.000	1.000
LOCAL	0.682	0.466	0.000	0.000	1.000	1.000	1.000
VOL	1.245	1.192	0.226	0.501	0.877	1.582	3.400
SIZE	20.832	0.966	19.350	20.207	20.809	21.410	22.439
LEV	0.470	3.656	0.167	0.351	0.481	0.634	1.503
STDROA	0.206	1.096	0.000	0.011	0.083	0.165	0.474
M/B	1.801	2.646	−2.566	1.055	1.825	2.899	5.418
INDN	5.346	1.316	3.295	4.263	6.182	6.548	6.681
INDSIZE	26.552	1.419	24.347	25.234	27.190	27.852	28.120

Table 2: descriptive statistics

6.2 Table 3 and Figure 1: correlations and nonlinearity

Read:

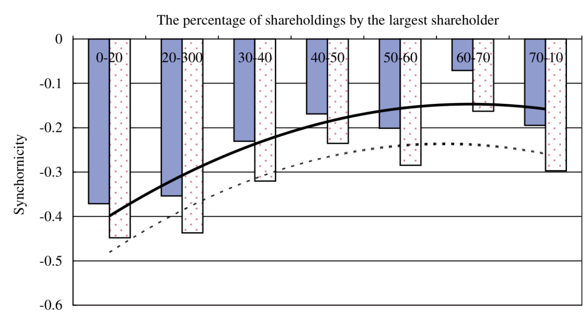
- Table 3 reports pairwise correlations among synchronicity, ownership, foreign ownership, audit quality, and controls.
- Synchronicity is positively correlated with largest shareholder ownership and government-related ownership.
- Synchronicity is negatively correlated with H-shares, B-shares, and Big 4 audits.
- Figure 1 plots synchronicity across bins of largest-shareholder ownership.
- Synchronicity rises until the largest shareholder owns roughly 40–50%, then begins to decline.

Economic message: The univariate evidence already suggests the paper’s main pattern: entrenchment dominates at lower concentration, but alignment dominates at high concentration.

Table 3
Correlation matrix.
All variables are as defined in the Appendix A. Superscripts a, b, and c stand for statistical significance at the 1%, 5%, and 10% levels, respectively.

	SYNCH (1*a*, *b*, *c*)	*TOP-HOLD*	*TOP-GOV*	*STATE*	*FORE-IGN*	*H-SHARE*	*B-SHARE*	*BIG4*
SYNCH(1)	0.971[a]	0.099[a]	0.066[a]	0.043[a]	−0.016	−0.011	−0.009	−0.041[a]
SYNCH (1*a*, *b*, *c*)	1	0.091[a]	0.066[a]	0.043[a]	−0.007	−0.018	−0.005	−0.037[a]
TOPHOLD		1	0.255[a]	0.459[a]	−0.018	0.091[a]	−0.047[a]	0.084[a]
TOPGOV			1	0.402[a]	0.048[a]	0.057[a]	0.015	0.033[b]
STATE				1	0.025[b]	0.064[a]	0.003	0.030[b]
FOREIGN					1	0.435[a]	0.639[a]	0.377[a]
HSHARE						1	−0.076[a]	0.273[a]
BSHARE							1	0.174[a]
BIG4								1
	LOCAL	*VOL*	*SIZE*	*LEV*	*STD-ROA*	*M/B*	*INDN*	*INDSIZE*
SYNCH(1)	0.002	−0.178[a]	0.134[a]	0.014	−0.003	−0.157[a]	−0.048[a]	−0.012
SYNCH (1*a*, *b*, *c*)	−0.001	−0.177[a]	0.108[a]	0.012	−0.007	−0.162[a]	−0.009	0.015
TOPHOLD	0.041[a]	−0.215[a]	0.184[a]	−0.020	0.049[a]	−0.111[a]	0.139[a]	0.158[a]
TOPGOV	0.013[a]	−0.195[a]	0.158[a]	0.024[c]	−0.012	−0.088[a]	0.122[a]	0.159[a]
STATE	0.016[a]	−0.137[a]	0.124[a]	0.027[b]	0.018	−0.058[a]	0.118[a]	0.142[a]
FOREIGN	−0.124[a]	−0.206[a]	0.159[a]	0.021	−0.026[c]	−0.276[a]	0.071[a]	0.064[a]
HSHARE	−0.250[a]	−0.129[a]	0.184[a]	0.004	−0.025[b]	−0.296[a]	0.081[a]	0.087[a]
BSHARE	−0.085[a]	−0.178[a]	0.048[a]	0.036[b]	−0.019	−0.207[a]	0.020	0.010
BIG4	−0.409[a]	−0.149[a]	−0.038[a]	0.118[a]	−0.021[c]	−0.458[a]	0.029[b]	0.047[a]
LOCAL	1	0.138[a]	0.098[a]	−0.104[a]	0.047[a]	0.375[a]	−0.033[b]	−0.070[a]
BIG4		−0.149[a]	−0.038[a]	0.118[a]	−0.021[c]	−0.458[a]	0.029[b]	0.047[a]
VOL		1	−0.206[a]	−0.040[a]	0.001	0.069[a]	−0.135[a]	−0.236[a]
SIZE			1	−0.235[a]	0.027[b]	−0.180[a]	0.041[a]	0.106[a]
LEV				1	0.076[a]	−0.061[a]	−0.051[a]	−0.046[a]
STDROA					1	0.041[a]	−0.119[a]	−0.119[a]
M/B						1	−0.015	−0.025[b]
INDN							1	0.953[a]

Table 3: correlation matrix



Shareholding (%)	0 ~ 20	20 ~30	30 ~ 40	40 ~ 50	50 ~ 60	60 ~ 70	70 ~ 100
SYNCH(1)	-0.371	-0.354	-0.230	-0.169	-0.201	-0.071	-0.195
SYNCH(1a,b,c)	-0.448	-0.437	-0.320	-0.235	-0.285	-0.163	-0.297
Number of observations (%)	515 (8.42)	1,369 (22.37)	995 (16.26)	968 (15.81)	1,075 (17.56)	777 (12.70)	421 (6.88)
Cumulative N (Cumulative %)	515 (8.42)	1,884 (30.79)	2,879 (47.05)	3,847 (62.86)	4,922 (80.42)	5,699 (93.12)	6,120 (100)

ween stock price synchronicity and the percentage of shareholdings by the largest shareholder The shaded ba
iile the dotted bar and the dotted curve represent *SYNCH*(1*a*, *b*, *c*).

Figure 1: synchronicity and largest shareholder ownership

6.3 Tables 4–5: baseline multivariate regressions

Read:

- Table 4 uses SYNCH(1) as the dependent variable.
- The coefficient on TOPHOLD^2 is negative and the coefficient on TOPHOLD is positive, supporting the concavity hypothesis.
- TOPGOV is positive and significant, meaning state-related largest shareholders increase synchronicity.
- HSHARE and BSHARE are negative, and the H-share coefficient is more negative.
- BIG44 is negative, consistent with high-quality auditors improving price informativeness.

- Table 5 repeats the analysis using SYNCH$(1a, b, c)$, and the results remain qualitatively similar.

Economic message: The baseline regressions show that ownership concentration, state control, foreign investor participation, and audit quality all matter for firm-specific information incorporation.

Table 4
The effect of ownership concentration, shareholding by foreign investors, and audit quality on stock price synchronicity.

All variables are as defined in the Appendix. The dependent variable is *SYNCH*(1), and is estimated using Eq. (1). Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscripts a, b, and c denote the 1%, 5%, and 10% levels of significance, respectively.

	(1a) H1a: $\gamma_1 < 0$ & $\gamma_2 > 0$ H1b: $\gamma_3 > 0$	(1b) H1a: $\gamma_1 < 0$ & $\gamma_2 > 0$ H1b: $\gamma_3 > 0$ (*TOPHOLD*=*DIF*)	(2) H2a: $\phi_1, \phi_2 < 0$ H2b: $\phi_1 < \phi_2 < 0$	(3) H3: $\lambda_1 < 0$	(4) Full-model regression
Panel A: Test variables					
TOPHOLD^2	−0.782	−0.012			−0.953
	(−2.12)[b]	(−2.12)[b]			(−2.52)[b]
TOPHOLD	0.899	0.152			0.981
	(2.61)[a]	(3.64)[a]			(2.80)[a]
TOPGOV	0.110	0.111			0.109
	(3.10)[a]	(3.14)[a]			(3.10)[a]
HSHARE			−0.405		−0.369
			(−4.81)[a]		(−4.22)[a]
BSHARE			−0.239		−0.215
			(−5.43)[a]		(−4.61)[a]
BIG4				−0.117	−0.059
				(−3.50)[a]	(−1.76)[c]
Panel B: Control variables					
LOCAL				0.111	0.082
				(3.45)[a]	(2.53)[b]
VOL	−0.076	−0.080	−0.117	−0.093	−0.111
	(−7.00)[a]	(−7.59)[a]	(−10.63)[a]	(−8.93)[a]	(−8.89)[a]
SIZE	0.074	0.069	0.091	0.065	0.075
	(4.53)[a]	(4.21)[a]	(5.69)[a]	(3.89)[a]	(4.44)[a]
LEV	0.001	0.001	0.003	0.003	0.003
	(0.43)	(0.32)	(1.19)	(1.06)	(1.32)
STDROA	−0.005	−0.006	−0.009	−0.004	−0.009
	(−0.47)	(−0.54)	(−0.73)	(−0.34)	(−0.81)
M/B	−0.030	−0.030	−0.045	−0.046	−0.050
	(−3.98)[a]	(−3.98)[a]	(−5.70)[a]	(−5.15)[a]	(−5.76)[a]
INDNUM	0.011	−0.004	−0.016	0.003	−0.041
	(0.09)	(−0.04)	(−0.13)	(0.02)	(−0.34)
INDSIZE	0.028	0.022	0.027	0.033	0.034
	(0.69)	(0.54)	(0.64)	(0.81)	(0.82)
Constant	−2.891	−2.404	−2.873	−2.680	−2.793
	(−3.02)[a]	(−2.47)[b]	(−2.95)[a]	(−2.78)[a]	(−2.89)[a]
Industry dummies	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included
N	6,120	6,088	6,120	6,120	6,120
Adj. R^2	34.61%	34.67%	35.15%	34.45%	35.95%

Table 4: baseline regressions using SYNCH(1)

Table 5
The effect of ownership concentration, shareholding by foreign investors, and audit quality on an alternative measure of stock price synchronicity.

All variables are as defined in the Appendix. The dependent variable is *SYNCH*(1*a*,*b*,*c*), and is estimated using Eqs. (1a), (1b), and (1c) for firms with A-shares only, firms with A- and B-shares, and firms with A- and H-shares, respectively. Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscripts a and b denote the 1% and 5% levels of significance, respectively.

	(1a) H1a: $\gamma_1 < 0$& $\gamma_2 > 0$ H1b: $\gamma_3 > 0$	(1b) H1a: $\gamma_1 < 0$ &$\gamma_2 > 0$ H1b: $\gamma_3 > 0$ (*TOPHOLD*=*DIF*)	(2) H2a: $\phi_1, \phi_2 < 0$ H2b: $\phi_1 < \phi_2 < 0$	(3) H3: $\lambda_1 < 0$	(4) Full-model regression
Panel A: Test variables					
TOPHOLD^2	−0.845	−0.011			−1.022
	(−2.23)[b]	(−2.02)[b]			(−2.66)[a]
TOPHOLD	0.916	0.148			1.013
	(2.60)[a]	(3.47)[a]			(2.85)[a]
TOPGOV	0.108	0.107			0.106
	(2.98)[a]	(2.93)[a]			(2.94)[a]
HSHARE			−0.444		−0.396
			(−5.19)[a]		(−4.48)[a]
BSHARE			−0.211		−0.186
			(−4.94)[a]		(−4.12)[a]
BIG4				−0.155	−0.095
				(−4.49)[a]	(−2.81)[a]
Panel B: Control variables					
LOCAL				0.125	0.094
				(3.67)[a]	(2.75)[a]
VOL	−0.076	−0.080	−0.115	−0.094	−0.110
	(−7.01)[a]	(−7.51)[a]	(−10.32)[a]	(−9.04)[a]	(−8.74)[a]
SIZE	0.058	0.052	0.074	0.044	0.056
	(3.26)[a]	(2.93)[a]	(4.35)[a]	(2.46)[b]	(3.11)[a]
LEV	0.001	−0.001	0.001	0.001	0.002
	(0.01)	(−0.08)	(0.53)	(0.51)	(0.72)
STDROA	−0.002	−0.003	−0.005	−0.001	−0.005
	(−0.20)	(−0.33)	(−0.52)	(−0.11)	(−0.56)
M/B	−0.034	−0.034	−0.049	−0.052	−0.057
	(−4.51)[a]	(−4.51)[a]	(−6.24)[a]	(−5.95)[a]	(−6.50)[a]
INDNUM	−0.045	−0.062	−0.070	−0.057	−0.097
	(−0.37)	(−0.51)	(−0.59)	(−0.47)	(−0.80)
INDSIZE	0.025	0.021	0.024	0.031	−0.032
	(0.64)	(0.53)	(0.61)	(0.81)	(−0.82)
Constant	−2.297	−1.833	−2.290	−2.012	2.190
	(−2.49)[b]	(−1.96)[b]	(−2.47)[b]	(−2.20)[b]	(2.37)[b]
Industry dummies	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included
N	6,120	6,088	6,120	6,120	5,994
Adj. R^2	35.56%	35.63%	36.11%	35.64%	36.91%

Table 5: alternative synchronicity measure

6.4 Tables 6–7: endogeneity and robustness

Read:

- Table 6 addresses endogeneity of Big 4 auditor choice using a first-stage auditor-choice model and two-stage regressions.
- The predicted Big 4 probability remains negatively related to synchronicity.
- Table 7 shows robustness to excluding 1997–1998, winsorizing outliers, and using weekly returns.
- The central results on ownership concentration, state ownership, foreign shares, and audit quality survive these checks.

Economic message: The robustness checks reduce concerns that the main findings are driven by the Asian financial crisis, outliers, return frequency, or endogenous auditor choice.

Table 6
Results of two-stage regressions to examine self-selection bias associated with auditor choice.

All variables except *SALES*, *SHRINC*, and *LOSS* are as defined in the Appendix. *SALES* represents the dollar amount of sales. *SHRINC* is an indicator variable that equal one if the number of shares outstanding increases more than 10% in each sample year and zero otherwise. *LOSS* is an indicator variable that equals one for loss firms and zero otherwise. *SYNCH* (1) is estimated using Eq. (1). *SYNCH*(1*a*,*b*,*c*) is estimated using Eqs. (1a)–(1c) for firms with A-shares only, firms with A- and B-shares, and firms with A- and H-shares, respectively. Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscripts a, b, and c denote the 1%, 5%, and 10% levels of significance, respectively.

	Section A First-stage probit regression	Section B Heckman approach with inverse Mills ratio (*Lamda*) included				Section C Two-stage least square (2SLS) approach with the predicted likelihood of auditor choice			
	(1) Dependent variable =*Pr(BIG4)*	(2) Dependent variable =*SYNCH*(1)		(3) Dependent variable =*SYNCH*(1*a*,*b*,*c*)		(4) Dependent variable =*SYNCH*(1)		(5) Dependent variable =*SYNCH*(1*a*,*b*,*c*)	
Panel A									
$TOPHOLD^2$			−0.992		−1.045		−0.738		−0.765
			$(-2.60)^{a}$		$(-2.71)^{a}$		$(-2.00)^{b}$		$(-2.04)^{b}$
TOPHOLD			1.012		1.031		0.828		0.827
			$(2.86)^{a}$		$(2.87)^{a}$		$(2.44)^{b}$		$(2.40)^{b}$
TOPGOV			0.111		0.106		0.093		0.085
			$(3.14)^{a}$		$(2.94)^{a}$		$(2.70)^{a}$		$(2.46)^{b}$
HSHARE			−0.384		−0.409		−0.263		−0.277
			$(-4.42)^{a}$		$(-4.64)^{a}$		$(-3.19)^{a}$		$(-3.33)^{a}$
BSHARE			−0.216		−0.187		−0.173		−0.141
			$(-4.70)^{a}$		$(-4.18)^{a}$		$(-3.82)^{a}$		$(-3.20)^{a}$
BIG4		−0.113	−0.057	−0.144	−0.088				
		$(-3.31)^{a}$	$(-1.72)^{c}$	$(-4.03)^{a}$	$(-2.55)^{b}$				
PredBIG4						−1.595	−1.412	−1.774	−1.593
						$(-8.83)^{a}$	$(-7.59)^{a}$	$(-9.91)^{a}$	$(-8.60)^{a}$
Panel B									
LOCAL		0.111	0.083	0.123	0.093	0.051	0.035	0.059	0.042
		$(3.38)^{a}$	$(2.49)^{b}$	$(3.52)^{a}$	$(2.65)^{a}$	(1.56)	(1.07)	$(1.71)^{c}$	(1.22)
VOL		−0.093	−0.110	−0.094	−0.110	−0.112	−0.121	−0.114	−0.122
		$(-8.98)^{a}$	$(-8.78)^{a}$	$(-9.12)^{a}$	$(-8.70)^{a}$	$(-10.15)^{a}$	$(-9.43)^{a}$	$(-10.36)^{a}$	$(-9.40)^{a}$
SIZE		0.064	0.075	0.043	0.057	0.033	0.041	0.009	0.019
		$(3.82)^{a}$	$(4.44)^{a}$	$(2.42)^{b}$	$(3.13)^{a}$	$(2.10)^{b}$	$(2.48)^{b}$	(0.56)	(1.08)
LEV	0.007	0.002	0.002	0.001	0.001	0.007	0.007	0.007	0.006
	$(8.05)^{a}$	(0.67)	(0.88)	(0.26)	(0.39)	$(1.85)^{c}$	$(1.96)^{c}$	(1.43)	(1.47)
STDROA		−0.004	−0.008	−0.001	−0.005	0.001	−0.004	0.004	0.001
		(−0.29)	(−0.70)	(−0.08)	(−0.45)	(0.04)	(−0.29)	(0.29)	(0.01)
M/B	−0.004	−0.052	−0.059	−0.058	−0.065	−0.117	−0.115	−0.130	−0.127
	$(-37.11)^{a}$	$(-4.96)^{a}$	$(-5.93)^{a}$	$(-5.53)^{a}$	$(-6.50)^{a}$	$(-9.65)^{a}$	$(-9.57)^{a}$	$(-10.71)^{a}$	$(-10.64)^{a}$
INDNUM		0.011	−0.033	−0.054	−0.094	−0.003	−0.036	−0.069	−0.098
		(0.09)	(−0.27)	(−0.45)	(−0.78)	(−0.02)	(−0.30)	(−0.58)	(−0.81)
INDSIZE		0.025	0.026	0.024	0.025	0.031	0.031	0.031	0.031
		(0.59)	(0.60)	(0.61)	(0.63)	(0.73)	(0.73)	(0.80)	(0.79)
SALES	0.001								
	$(4.53)^{a}$								
SHRINC	*−0.011*								
	(−1.52)								
Lamda		−0.007	−0.008	−0.009	−0.009				
		(−1.33)	(−1.53)	(−1.55)	$(-1.69)^{c}$				
LOSS	−0.054								
	$(-6.93)^{a}$								
Constant	0.131	−2.492	−2.635	−1.850	−2.053	−1.684	−1.808	−0.963	−1.140
	$(11.30)^{a}$	$(-2.55)^{b}$	$(-2.68)^{a}$	$(-2.00)^{b}$	$(-2.19)^{b}$	$(-1.73)^{c}$	$(-1.84)^{c}$	(−1.05)	(−1.22)
Industry dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included	Included	Included	Included	Included
N	5,994	5,994	5,994	5,994	5,994	5,994	5,994	5,994	5,994
Adj. R^2 (pseudo R^2)	(23.72%)	34.87%	36.43%	36.09%	37.40%	36.98%	38.02%	38.43%	39.22%

Table 6: Big 4 endogeneity checks

Table 7
The results of other robustness checks.

All variables are as defined in the Appendix A. The dependent variable is *SYNCH*(1), and is estimated using Eq. (1). Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscripts a, b, and c denote the 1%, 5%, and 10% levels of significance, respectively.

	Section A Excluding 1997 & 1998 observations		Section B Winsorising top & bottom 1%		Section C Use of weekly returns	
	(1a)	(1b)	(2a)	(2b)	(3a)	(3b)
Panel A: Test variables						
$TOPHOLD^2$	−1.106	−1.141	−1.349	−1.212	−1.348	−1.405
	$(-2.65)^{a}$	$(-2.72)^{a}$	$(-2.82)^{a}$	$(-2.73)^{a}$	$(-2.78)^{a}$	$(-2.87)^{a}$
TOPHOLD	1.177	1.207	1.261	1.074	1.137	1.184
	$(3.01)^{a}$	$(3.07)^{a}$	$(2.91)^{a}$	$(2.67)^{a}$	$(2.58)^{b}$	$(2.67)^{a}$
TOPGOV	0.121	0.123	0.085	0.084	0.110	0.117
	$(3.05)^{a}$	$(3.07)^{a}$	$(2.77)^{a}$	$(2.77)^{a}$	$(2.53)^{b}$	$(2.67)^{a}$
HSHARE	−0.388	−0.408	−0.391	−0.404	−0.246	−0.267
	$(-3.94)^{a}$	$(-4.14)^{a}$	$(-4.61)^{a}$	$(-4.79)^{a}$	$(-2.43)^{b}$	$(-2.63)^{a}$
BSHARE	−0.229	−0.230	−0.210	−0.181	−0.196	−0.199
	$(-4.45)^{a}$	$(-4.52)^{a}$	$(-5.01)^{a}$	$(-4.53)^{a}$	$(-3.77)^{a}$	$(-3.89)^{a}$
BIG4	−0.071	−0.071	−0.097	−0.099	−0.083	−0.078
	$(-1.90)^{c}$	$(-1.91)^{c}$	$(-3.99)^{a}$	$(-4.11)^{a}$	$(-2.09)^{b}$	$(-1.93)^{c}$
Panel B: Control variables						
LOCAL	0.085	0.087	0.102	0.099	0.100	0.097
	$(2.41)^{b}$	$(2.42)^{b}$	$(3.60)^{a}$	$(3.49)^{a}$	$(2.43)^{b}$	$(2.31)^{b}$
VOL	−0.123	−0.122	−0.180	−0.185	−0.128	−0.127
	$(-8.64)^{a}$	$(-8.53)^{a}$	$(-11.89)^{a}$	$(-12.39)^{a}$	$(-8.51)^{a}$	$(-8.39)^{a}$
SIZE	0.072	0.072	0.028	0.005	−0.004	−0.002
	$(3.89)^{a}$	$(3.87)^{a}$	$(1.72)^{c}$	(0.33)	(−0.16)	(−0.11)
LEV	0.003	0.002	−0.359	−0.367	0.002	0.002
	(1.35)	(0.91)	$(-7.33)^{a}$	$(-7.43)^{a}$	(1.02)	(0.72)
STDROA	−0.013	−0.012	0.250	0.252	−0.007	−0.006
	(−0.96)	(−0.82)	$(2.83)^{a}$	$(2.95)^{a}$	(−0.60)	(−0.44)
M/B	−0.053	−0.064	−0.101	−0.104	−0.050	−0.061
	$(-5.48)^{a}$	$(-5.82)^{a}$	$(-10.32)^{a}$	$(-9.90)^{a}$	$(-4.60)^{a}$	$(-4.85)^{a}$
INDNUM	−0.002	0.002	−0.240	−0.267	−0.101	−0.106
	(−0.12)	(0.01)	$(-2.35)^{b}$	$(-2.67)^{a}$	(−0.66)	(−0.68)
INDSIZE	−0.005	−0.016	−0.021	0.001	0.073	0.067
	(−0.12)	(−0.36)	(−0.50)	(0.04)	(1.42)	(1.27)
Lamda		−0.010		−0.008		−0.012
		(−1.63)		$(-1.73)^{c}$		$(-1.78)^{c}$
Constant	−1.107	−0.891	0.872	0.841	−1.607	−1.494
	(−1.19)	$(-0.93)^{a}$	(0.87)	(0.89)	(−1.35)	(−1.23)
Industry dummies	Included	Included	Included	Included	Included	Included
Year dummies	Included	Included	Included	Included	Included	Included
N	5,241	5,120	6,120	5,994	6,080	5,955
Adj. R^2	34.90%	35.36%	40.38%	43.38%	20.99%	21.45%

Table 7: robustness checks

6.5 Table 8: earnings informativeness validation

Read:

- Table 8 examines whether high-synchronicity firms have weaker return-earnings associations.
- The interaction $NI \times DR\_SYNCH$ is negative and significant under both synchronicity measures.
- The main earnings coefficient is positive, so earnings are value relevant on average.
- The negative interaction means earnings are less strongly capitalized into prices when synchronicity is high.

Economic message: Table 8 validates the interpretation of synchronicity: high synchronicity means stock prices contain less firm-specific earnings information.

Table 8
The effect of stock price synchronicity on return-earnings associations.

All variables are as defined in the Appendix. *DR_SYNCH*(1) and *DR_SYNCH*(1*a*, *b*, *c*) are the scaled decile rank scores of *SYNCH*(1) and *SYNCH*(1*a*, *b*, *c*), respectively, which range from zero to one. *SYNCH*(1) is the stock price synchronicity measure, which is estimated using Eq. (1), while *SYNCH*(1*a*, *b*, *c*) is the stock price synchronicity measure, which is estimated using Eqs. (1a), (1b), and (1c) for firms with A-shares only, firms with A- and B-shares, and firms with A- and H-shares, respectively. Numbers in parentheses represent *t*-values that are adjusted using standard errors corrected for clustering at the firm level. The superscript a denotes the 1% level of significance, respectively.

	(1) Using *DR_SYNCH*(1)	(2) Using *DR_SYNCH*(1*a*, *b*, *c*)
NI	2.894	2.984
	(4.83)[a]	(4.78)[a]
*NI*DR_SYNCH*	−0.271	−0.340
	(−3.22)[a]	(−3.72)[a]
*NI*MCAP*	0.127	0.130
	(5.00)[a]	(4.99)[a]
*NI*LEV*	−0.012	−0.015
	(−0.46)	(−0.55)
*NI*M/B*	0.086	0.086
	(3.71)[a]	(3.49)[a]
CONSTANT	−0.046	−0.046
	(−1.53)	(−1.54)
Year dummies	Included	Included
Industry dummies	Included	Included
Adj. R^2	5.11%	5.26%
N	4,654	4,654
F	10.85 ($p < 0.0001$)	11.19 ($p < 0.0001$)

7 Short summary

- The paper studies Chinese listed firms from 1996–2003.
- It asks whether ownership concentration, state ownership, foreign shareholding, and audit quality affect stock price synchronicity.
- Stock price synchronicity is measured as $\log(R^2/(1-R^2))$ from market-model regressions.
- Synchronicity is concave in largest-shareholder ownership and peaks around 50% ownership.
- Government-related largest shareholders increase synchronicity.
- H-share and B-share firms have lower synchronicity, with stronger effects for H-share firms.
- Big 4 auditors reduce synchronicity.
- Robustness checks address auditor-choice endogeneity, the Asian financial crisis, outliers, and return-frequency choices.
- Earnings are less strongly reflected in returns when synchronicity is high.
- The central lesson is that firm-level governance affects price informativeness even within a weak investor-protection environment.

8 Conclusion

8.1 Core conclusion

Gul, Kim, and Qiu show that firm-level governance characteristics influence the amount of firm-specific information incorporated into Chinese stock prices. Stock price synchronicity is higher when ownership concentration is in the entrenchment range and when the largest shareholder is government-related, but lower when foreign investor participation and Big 4 auditing improve monitoring and disclosure.

8.2 Economic intuition

Large shareholders can either hide information to protect private benefits or reveal information to protect their investment value. Foreign investors and Big 4 auditors create pressure for credible disclosure. The Chinese setting allows these forces to be studied side by side within one emerging market.

8.3 Takeaway

The paper's main message is that stock prices become more informative when governance mechanisms reduce controlling-shareholder entrenchment and increase credible disclosure. Improving price informativeness in emerging markets therefore requires both country-level investor protection and firm-level governance reforms.